DOI: 10.1111/jdv.12152 JEADV

ORIGINAL ARTICLE

Management of advanced non-melanoma skin cancers using helical tomotherapy

N. Kramkimel,[1,2,*] R. Dendale,[1] S. Bolle,[1] S. Zefkili,[1] A. Fourquet,[1] Y.M. Kirova[1]

[1]Department of Radiation Oncology, Curie Institute, Paris, France
[2]Department of Dermatology, Cochin Hospital, Paris, France
*Correspondence: N. Kramkimel. E-mail: nora.kramkimel@cch.aphp.fr

Abstract

Background Helical tomotherapy (HT) is a relatively new method of radiotherapy, the main advantages of which are an increase of irradiation dose on the target tumour volume and best protection of adjacent organs at risk.

Objective To provide an accurate evaluation of efficiency and tolerance of HT on different kinds of non-melanoma skin tumours.

Patients and methods We analysed, retrospectively, 25 patients (pts) who were treated with HT for advanced non-melanoma skin cancers. We studied the characteristics of patients and tumours, associated treatments, characteristics, efficacy and tolerance of HT treatment.

Results Eight had basal cell carcinoma, Eight had cutaneous squamous cell carcinoma, five Merkel cell carcinoma and four adnexal carcinoma. The median age was 75 years (range 51–89). The median follow-up was 12 months. HT was used because of incomplete excision (n = 12), lymph node involvement (n = 6), non-operable lesions (n = 4) and high risk of relapse (n = 4). The delivered dose for the tumour bed was between 50 and 70 Gy and for the lymph node it was between 50 and 64 Gy. There was no grade III–IV adverse event, except one grade III mucositis. Fourteen pts suffered from limited toxic effects 6 months after the end of the treatment, mainly loss of eyebrow and teared eye. Complete remission was noticed for 22 pts (88%); one pt had progressive disease and two pts died.

Conclusion HT is an effective option for advanced non-melanoma skin cancers as radical treatment or in association with surgery. Early and late toxicity and cosmetic results are acceptable.

Received: 23 December 2012; Accepted: 6 March 2013

Conflict of interest

The authors declare no conflict of interest.

Funding sources

None.

Introduction

Radiotherapy plays an important role in the treatment of skin tumours such as those arising from non-melanoma skin cancer and primary cutaneous lymphoma. For non-melanoma skin cancers, including basal cell carcinoma (BCC), cutaneous squamous cell carcinoma (CSC), Merkel cell carcinoma (MCC), and adnexal carcinoma (AC), complete surgical excision with a security margin is still the reference therapy, but the choice of the treatment technique depends on various factors such as the size of the tumour, the general condition of the patient, cosmetic considerations, and recurrence. Radiotherapy is an effective option for tumour control and quality of life improvement. Histological confirmation is essential before radiotherapy initiation.

A part of this work was presented as poster at the ASTRO 54th Annual Meeting, Boston, Oct 28–30, 2012 and as oral presentation at the Advancing Radiation Oncology Forum in Dallas, Nov 28–30, 2012, USA.

For BCCs, radiotherapy is an option when surgery is not possible in such circumstances as recurrence, incomplete excision, greater than 2-cm nodular disease in the head, or bone or cartilage involvement.[1] In these cases, 5–10-mm security margins are observed depending on the tumour size and superficial X-rays and electron beams are usually employed. A randomized trial comparing radiotherapy to surgical excision of facial BCC of less than 4-cm diameter found 4-year failure rates of 7.5% and 0.7%, respectively (P = 0.003). Cosmetic outcome also significantly favored surgical excision at 4 years, with 87% of the surgery-treated patients and 69% of the radiation-treated patients finding the cosmetic results satisfactory (P < 0.01). The study used different techniques of radiotherapy including interstitial brachytherapy, contact therapy, and conventional radiotherapy.[2] Despite these results, radiotherapy is still an option for BCCs that occur in areas where surgery would be technically difficult or would result in unacceptable tissue destruction, or for patients who would not be able to tolerate surgery.[3] Efficacy of electron beam radiotherapy for BCCs was evaluated and the 3-year local recurrence-free survival rates were 97.6% for tumours treated with 54 Gy in 18 fractions and 96.9% with 44 Gy in 10 fractions.[4] Radiotherapy has also been compared to imiquimod 5% cream for nodular BCCs on the eyelid; in this study, remission rates were equivalent in the two groups, but tolerability was better within the radiotherapy group during the treatment.[5]

For CSCs, radiotherapy is recommended when surgery is not possible or used as adjuvant in treatment of high-risk tumours with 10–15-mm security margins.[6] Three-year freedom from local recurrence was 97% with 54 Gy in 18 fractions and 93.6% with 44 Gy in 10 fractions in a study about electron beam therapy for CSCs.[4] Radiotherapy is particularly an option for CSCs overlying cartilage,[7] those with perineural invasion,[8,9] or with recurrence.[10] Adjuvant radiotherapy is also indicated on regional lymph node involvement.[11,12]

For MCCs, radiotherapy is applied to the cavity of the excised primary or recurrent tumour.[13] Adjuvant regional radiotherapy on tumour bed decreases regional recurrence but has no benefit on overall survival.[14,15] Radiotherapy alone can be a reasonable option for inoperable MCCs with similar overall and disease-free survival rates to combination of surgery and radiotherapy.[16] A recent study showed that radiation monotherapy could also be used for lymph node-positive MCC (for both microscopic and palpable lymph node disease), with comparable results to lymphadenectomy with or without adjunct radiotherapy.[17,18]

As regards ACs, few cases have been published about radiotherapy for sebaceous carcinoma. Sebaceous carcinomas have been considered radioresistant for years. But recently, reports of success with radiotherapy for sebaceous carcinoma have been reported. Local control at the primary site after radiation and/or surgery was 90% in a study including 10 patients.[19] An appropriate radiotherapy delivery regimen (>55 Gy) seems to be an effective treatment for sebaceous carcinoma in combination or as an alternative to surgery.[20,21]

The choice of the most appropriate irradiation technique should depend on tumour localization and clinical target volume (CTV), as much as histology, and the size and thickness of the tumour. The three standard radiation techniques used for skin cancers are low energy X-rays for tumours with less than 1-cm thickness, electron beams for thicker ones, and brachytherapy for lesions of lower lip and nasal pyramid. Helical TomoTherapy® (Accuray Incorporated, Sunnyvale, CA, USA) is a new technique of radiotherapy and little data has been published on non-melanoma skin cancer. Helical Tomotherapy® is an intensity-modulated radiation treatment using 6 MV photons combined with megavoltage CT imaging allowing an accurate patient positioning. This dynamic therapy has been developed at the University of Wisconsin in the late 1980s.[22] The system is mounted upon a continuous rotating gantry coupled with translation of the patient through the gantry. The helical delivery minimizes the risk of significant high or low dose deposition in areas of overlap or junctioning. It is adaptative radiotherapy with constant possibility to verify therapeutic plan: patient positioning, target tumour/organ registration and reconstruction of delivered dose. The dose is delivered slice by slice during an axial computed tomography scan.[23] The main advantages are increase of irradiation dose on the target tumour volume and best protection of adjacent organs at risk.[24] Indeed, this technique allows targeting areas of disease while simultaneously minimizing dose to organ at risk.[25] It is useful for large size tumour, complex form tumour and when organs at risk are close or inside the tumour. Head and neck cancer is one of the best indications for tomotherapy. It has also already been used against sarcomas, gynecological and lung tumours and hematologic diseases.[26–28] Helical tomotherapy achieved better target coverage in head and neck cancer with better protection of organs at risk and reduction of xerostomia compared to step-and-shoot intensity-modulated radiation treatment.[29] Less than 10 hospitals possess helical tomotherapy alternative in France, and Curie Institute is the only one in Paris since 2007. A second machine arrived in 2010 at Curie Institute.

The aim of this study was to provide an evaluation of the efficacy and tolerance, including early and late cosmetic results, of helical tomotherapy for different kinds of non-melanoma skin tumours.

Patients and methods

We analysed online medical records of patients treated with Helical Tomotherapy for skin lesions at the Radiation Oncology department of Curie Institute (Paris, France) between 2007 and 2010.

The inclusion criterion was a biopsy-confirmed skin tumour of the basal cell carcinoma, cutaneous squamous cell carcinoma, Merkel cell carcinoma, or adnexal carcinoma histology.

Patient (age, sex) and tumour (histology, site, primary or recurrent nature, lymph node involvement, visceral metastasis) characteristics, the nature of any adjuvant treatments (surgery, lymph node dissection, chemotherapy), and characteristics of the TomoTherapy treatment [planning target volume (PTV), total dose, fractions, organs at risk (OAR) doses] were studied. The results of the treatment as far as efficacy (follow-up duration, recurrence, remission, death) and toxicity (early and late cosmetic results corresponding to immediate and at 6-month follow-ups) were recorded. Complete response (CR) signified disappearance of the lesion during follow-up,and progressive disease any enlargement of the lesion or local or remote recurrence. All toxicities were described using the Common Terminology Criteria for Adverse Events (CTCAE), version 3.0 (www.eortc.be/services/doc/ctc/ctcaev3.pdf). Clinical assessment was

Table 1 Basal cell carcinomas: characteristics of patients and tumours, tomotherapy: technic data, response and tolerance

	Sex	Age (years)	Primary/ recurrent tumour	Histological type of CBC	Localisation	Surgery	PTV	Total dose (Gy) (fractions)	Early adverse events	Follow-up/ Response	Late adverse events
1	M	79	2nd recurrence	Infiltrant	Right forehead	Incomplete excision, bone infiltrat	CTV	66.6 (1.8)	Radiodermatitis grade I Localized alopecia	31 months/CR	Loss of right temporal eyebrow
2	M	51	Recurrent	Infiltrant	Left forehead	Complete excision	Left forehead	54 (2)	Radiodermatitis grade II	12 months/CR	Left upper lid oedema
3	F	73	Primary	Infiltrant	Left forehead	Incomplete excision, bone and meningeal infiltrat, lymph node involvment (1N+R+/11N)	Tumour and left cervical lymph node Metastatic lymph node	52 (1.8) 64 (2.2)	Radiodermatitis grade II Mucositis grade I	18 months/CR	0
4	F	73	Primary	Sclerosis	Right nasal canthus with eye invasion	Not possible	Right eye-socket and nose root	66 (2.13)	Radiodermatitis grade II	8 months/CR	Right teared eye
5	F	76	Primary	Sclerosis	Right upper lid	Not possible	Right eye-socket	66 (2)	Radiodermatitis grade II Conjunctival hyperhemia	10 months/CR	Loss of right eyebrow
6	M	61	Numerous recurrences	Sclerosis	Left nasal canthus and left lower lid	Incomplete excision	Left nasal canthus and left lower lid	66 (2)	Radiodermatitis grade I	11 months/CR	Loss of left eyebrow Left teared eye
7	M	83	Primary	Sclerosis	Left lower lid	Incomplete excision	Left lower lid, left eye-socket, left nasal canthus	66 (2)	Radiodermatitis grade I	2 months/CR	
8	M	64	Recurrent	Sclerosis	Left lower lid	Incomplete excision	Left lower lid	66 (2)	Radiodermatitis grade I Left teared eye	2 months/CR	

Table 2 Cutaneous squamous cell carcinomas: characteristics of patients and tumours, tomotherapy: technic data, response and tolerance

	Sex	Age (years)	Primary/ recurrent tumour	Localisation	Surgery	N	TNM/Stage*	PTV	Total dose (Gy) (fractions)	Early adverse events	Follow-up / Response	Late adverse events
1	F	64	Recurrent	Right nasal canthus	Complete excision	N0	T2N0M0/II	Right nasal canthus	56 (2)	Conjunctival hyperemia Radiodermatitis grade I	6 months/CR	0
2	M	83	Primary	Right lower lid	Complete excision	N0	T2N0M0/II	First right cervical lymph node	50 (2)	Radiodermatits grade II Chondritis	18 months/ CR	0
3	M	80	Primary	Right temple	Complete excision	N0	T2N0M0/II	Tumour Right parotid/ cervical lymph node II	60 (2) 54 (1.8)	Radiodermatitis grade I	9 months/CR	0
4	M	86	Primary	Left forehead	Complete excision	N+ (3N + /19N)	T2N1bM0/III	Left cervical lymph node II–III Left cervical lymph node IV-V and pretragis	59.4 (1.66) 54 (1.63)	Radiodermatitis grade II	5 months/CR	
5	M	78	Primary	Left eye-socket	Incomplete exenteration	N0	T4N0M0/III	Left eye-socket and left cervical lymph node	50 (2)	Radiodermatitis grade II Hypoacusia Alopecia	39 months/CR	Left nasal canthus itching
6	M	85	Primary	Left eye	Not possible	N0	T4N0M0/III	Left eye Left cervical lymph node and left eye-socket	66 (2.35) 50.4 (1.8)	Milphosis Conjunctival hyperemia Palpebral oedema	5 months/Death	
7	M	60	Primary	Right nasal canthus and conjunctiva	Incomplete excision	N0	T3N0M0/II	CTV Right eye Right cervical lymph node	54 (1.74) 66 (2.13) 54 (1.74)	Alopecia Conjunctival hyperemia Mucositis grade I	24 months/CR	Milphosis
8	F	89	Primary	Right eye	Not possible	N0	T4N0M0/III	Right eye Right eye-socket First cervical lymph node	66 (2.35) 54.9 (1.96) 50.4 (1.8)	Radiodermatitis grade I Mucositis grade I	24 months/CR	Milphosis Hypoacusia Palpebral synechia

*AJCC Cancer Staging Manual. 7th ed.

Table 3 Merkel cell carcinomas: characteristics of patients and tumours, tomotherapy: technic data, response and tolerance

	Sex	Age (y)	Primary or recurrent tumour	Localisation	Surgery	N	M	TNM/Stage*	PTV	Total dose (Gy) (fractions)	Early adverse events	Follow-up / Response	Late adverse events	Chemotherapy
1	F	73	Primary	Right cheek	Incomplete depth excision	N0	M0	T2pN0M0/IIa	Tumour	66 (2)	Radiodermatitis grade II Mucositis grade II Dysphagia grade II Conjunctival hyperhemia	7 months/CR	0	Vepeside (neoadjuvant)
2	F	73	Primary	Right cheek	Complete excision	2N+R +/21N	M0	T3N1bM0/IIIb	Tumour and right cervical lymph node Metastatic lymph node	52 (1.8) 64 (2.2)	Radiodermatitis grade II Mucositis grade II Dysphagia grade II Localized alopecia	20 months/CR	Right teared eye Limitation of oral cavity opening	0
3	M	70	Primary	Left parietal scalp	Complete excision	3N+2R +/24N	M0	T2N1bM0/IIIb	Tumour and left cervical, susclavicular and pretargis lymph node and left parotid Metastatic lymph node	54 (1.8) 60 (2)	Radiodermatitis grade I Mucositis grade I	9 months/CR	0	0
4	M	87	Primary	Near left eye	Complete excision	Parotid 12N+/15N Lymph node 9N+/20N	M0	T3N1bM0/IIIb	Right cheek Parotid and metastatic lymph node Cervical lymph node	65/53 (2.1/1.7) 65 (2.1) 60 (1.93)	Radiodermatitis grade II Mucositis grade III Alopecia	14 months/ progressive disease (visceral metastasis)	Cutaneous pigmentation -Xerostomia	0
5	M	81	Metastatic right cervical lymph node recurrence	Forehead	Lymph node dissection	5N+R+/6N	M0	T2N1bM0/IIIb	Right cervical lymph node and parotid Tumour	56 (1.7) 66 (2)	Radiodermatitis grade I Mucositis grade II Dysphagia grade II	12 months/CR	Cutaneous dystrophia	Etoposide

*AJCC Cancer Staging Manual. 7th ed.

Table 4 Adnexal carcinomas: characteristics of patients and tumours, tomotherapy: technic data, response and tolerance

	Sex	Age (years)	Primary/ recurrent tumour	Histological type	Localisation	Surgery	N	TNM/Stage*	PTV	Total dose (Gy) (fractions)	Early adverse events	Follow-up / Response	Late adverse events
1	F	68	Recurrence	Infiltrant adnexal carcinoma	Right lower lid	Incomplete excision	N0	T1N0M0/I	Right lower lid Right cervical lymph node and parotid	64 (2.28) 50 (1.78)	Radiodermatitis grade I Loss of eyelash Conjunctival hyperhemia	22 months/CR	Right teared eye
2	M	77	Primary	Sebaceous carcinoma	Left eye-socket and nose root	Incomplete excision	N0	T2N0M0/II	Left eye-socket and nose root	66 (2.2)	Radiodermatitis grade II	7 months/ progressive disease (local recurrence and death)	Left cheek oedema
3	M	61	Primary	Unclassifiable adnexal carcinoma	Left lower lid	Incomplete excision	N0	T2N0M0/II	Left lower lid Left cervical lymph node	65 (2.22) 50 (1.85)	Radiodermatitis grade I Loss of eyelash Conjunctival hyperhemia	14 months/CR	Left teared eye
4	F	75	Primary	Mucoepidermoid carcinoma	Right lacrimal canal	Incomplete excision	N0	T2N0M0/II	Right eye-socket Right cervical lymph node Right parotid	70/59 (2.12/1.78) 54 (1.63) 59 (1.78)	Radiodermatitis grade I Alopecia	18 months/CR	0

*AJCC Cancer Staging Manual. 7th ed.

Table 5 Mean doses received by OAR (Gy) according to the localization of the tumour: eye area (a), forehead (b), cheek (c)

(a)

Right optic nerve			Left optic nerve			Right eye			Left eye			Right lens			Left lens			Right cochlea			Left cochlea			Oral cavity			Brain stem			Pituitary gland		
Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max
12.6	19	25	15.4	26	34.3	10.3	17.4	26.9	12.1	29.1	51	4.4	5.2	8.6	17.5	27	40.9	13.6	15.4	17.3	10	11.5	12.7	6.8	15.5	42.2	5.1	11.5	21.4	11.7	16.4	23.9

(b)

Right optic nerve			Left optic nerve			Right eye			Left eye			Optic chiasm			Right cochlea			Left cochlea			Brain stem			Pituitary gland			Spinal cord		
Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max
9.6	15.2	21.7	2.6	7.2	10.8	5.1	9.8	26.1	1.2	4.8	13.7	11.4	14	17.2	7.2	9.1	13.8	8.9	11.3	15	3.4	8.9	20	11.8	14.5	17.1	2.4	10.8	25.1

(c)

Right optic nerve			Optic chiasm			Right cochlea			Brain stem			Spinal cord		
Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max
12.2	17.1	22.2	8.6	10.8	12.7	6.6	9.3	14.5	5.4	10.3	15.4	0.8	6.4	16.7

Min, minimum; Avg, average; Max, maximum.

performed weekly during radiotherapy and, after completion of radiotherapy, every 1 or 2 months, then once a year.

For statistical analysis, quantitative data was expressed as mean ± SD, and qualitative data as frequency and percentage.

Results

Twenty-five patients were included in the study. Eight (33.3%) had BCC, eight (33.3%) CSC, five (20.8%) MCC and four (16.6%) AC. There were 9 women (36%) and 16 men (64%) with the male to female ratio of 1.77. Mean age was 74 years (range, 51–89 years; median 75 years). All tumours were facial.

Helical TomoTherapy was indicated because of incomplete excision (n = 12), lymph node involvement (n = 6), non-operable lesions (n = 4) and high risk of relapse (n = 4). The PTV was tumour bed and/or the regional lymph nodes with 1-cm margins. Delivered dose for a tumour bed was between 50–70 Gy and for lymph nodes between 50–64 Gy.

Mean follow-up was 13.7 months (range, 2–39 months; median, 12 months). CR was noted for 22 patients (88%). One patient with MCC had a visceral metastasis at 14-month follow-up. Two patients died (8%): one with inoperable CSC at 5 months and the other with recurrent AC at 7 months.

There were no grade IV adverse events, and only one case of grade III mucositis. Grade I or II early radiodermatitis was diagnosed respectively in 12 and 11 patients. Late adverse events were analysed for 21 patients. Fourteen of the 21 patients (66.6%) suffered from limited toxic effects 6 months after the end of the treatment, mainly in the form of loss of eyebrow or hyperlacrimation.

Patient and tumour characteristics, tomotherapy data, response and tolerance are summarized in the following tables for each pathology (Tables 1–4).

The dose levels (minimum, mean, maximum) received by OAR for each tumour site are reported in Table 5.

Example of dose distribution is given in Fig. 1. Clinical aspect before and after tomotherapy of an incomplete excised BCC of forehead and nasal canthus is showed in Figs 2–4.

Discussion

To our knowledge, this study is the largest cohort analysis of Helical TomoTherapy treatment of advanced non-melanoma skin cancer. Currently, only few case-reports have been published about the efficacy of TomoTherapy in skin cancer: one case of multiple BCCs treated simultaneously in the same patient,[30] one case of extensive BCC of the forehead and the scalp,[31] one case of sarcomatoid sarcoma of the scalp,[32] one case of recurrent eccrine mucinous adenocarcinoma of the scalp,[33] and three cases of skin malignancies without detail.[34]

In our study, at a median 12-month follow-up, complete remission was observed in 88% of tumours: 100% in BCCs, 85.7% in CSCs, 80% in MCCs, and 75% in ACs. This is a promising response rate especially because our patients suffered from

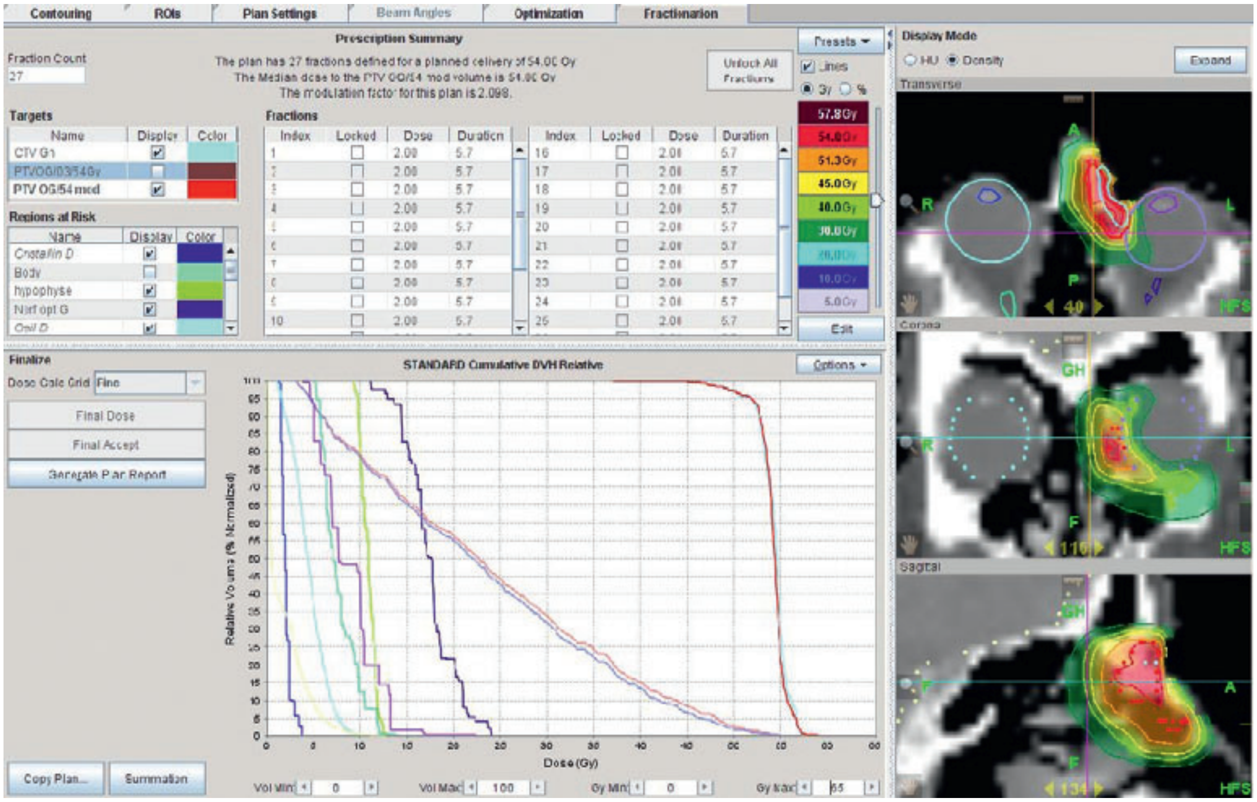

Figure 1 Dose distribution and DVH for non-operable basal cell carcinoma.

Figure 2 Incomplete excision of a basal-cell carcinoma of the forehead and nasal canthus: after surgery and before tomotherapy.

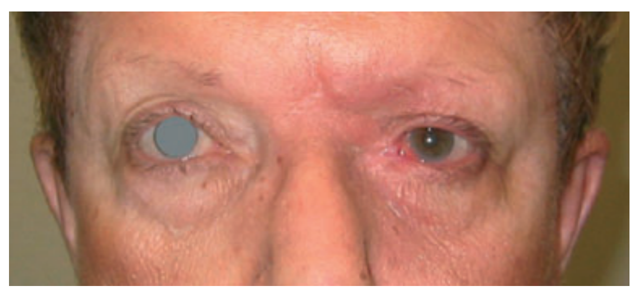

Figure 3 Incomplete excision of a basal-cell carcinoma of the forehead and nasal canthus: at the end of tomotherapy.

hard skin lesions; indeed, tumours treated in this study had poor prognoses either because of their localization (face), incomplete surgical excision, recurrent or inoperable nature, or lymph node involvement.

As regards toxicity, only one grade III adverse event was recorded (mucositis) in a patient who recevied 65 Gy to the right cheek and parotid. Other early adverse events were grades I or II (mainly radiodermatitis, mucositis, conjunctival hyperemia, alopecia, and dysphagia) and were in accord with reports on other forms of radiotherapy. Late adverse events, signifying events occurring at 6 months or later post TomoTherapy treatment, presented in 14 patients. Two types of late adverse event were recognized: cosmetic adverse events (loss of an eyebrow or eye lashes, disfigurement of the skin, or edema) and functional adverse events (hyperlacrimation, palpebral synechia, difficulty opening mouth); the adverse events were generally tolerable in all cases.

Average doses received by OARs were under respective limits of tolerance.

Figure 4 Incomplete excision of a basal-cell carcinoma of the forehead and nasal canthus: 3 months after the end of tomotherapy.

Our study has its limitations: it is a retrospective single-center study based on online medical records of a small number of patients. Its results need to be confirmed by larger prospective studies with longer follow-up and possibly including comparisons between TomoTherapy and other radiotherapy techniques.

Nevertheless, the results are encouraging in terms of toxicity with good sparing of the OAR.

Conclusion

Helical TomoTherapy appears to be an effective option of radical treatment for advanced non-melanoma skin cancers as monotherapy or in combination with surgery. Early, late toxicity and cosmetic results were satisfactory. The method facilitates sparing of important OARs. These results should be confirmed by larger prospective studies with longer follow-up.

Acknowledgments

Thanks to Mikail Gezginci, PhD, of Accuray for reviewing this work for language; and to Malika Amessis, Dominique Peurien and Chantal Dauphinot for their help in the dosimetric work.

References

1 Coulomb A. Recommendations for basal cell carcinoma. *Ann Dermatol Venereol* 2004; **131**: 661–756.
2 Avril MF, Auperin A, Margulis A *et al.* Basal cell carcinoma of the face: surgery or radiotherapy? Results of a randomized study. *Br J Cancer* 1997; **76**: 100–106.
3 Smith V, Walton S. Treatment of facial Basal cell carcinoma: a review. *J Skin Cancer Epub* 2011; **2011**: 380371.
4 Van Hezewijk M, Creutzberg CL, Putter H *et al.* Efficacy of a hypofractionated schedule in electron beam radiotherapy for epithelial skin cancer: analysis of 434 cases. *Radiother Oncol* 2010; **95**: 245–249.
5 Garcia-Martin E, Gil-Arribas LM, Idoipe M *et al.* Comparison of imiquimod 5% cream versus radiotherapy as treatment for eyelid basal cell carcinoma. *Br J Ophthalmol* 2011; **95**: 1393–1396.
6 French Society of Dermatology. Guideline for the diagnosis and treatment of cutaneous squamous cell carcinoma and precursor lesions. *Ann Dermatol Venereol* 2009; **136**(Suppl 5): S166–S186.
7 Caccialanza M, Piccinno R, Percivalle S, Rozza M. Radiotherapy of carcinomas of the skin overlying the cartilage of the nose: our experience in 671 lesions. *J Eur Acad Dermatol Venereol* 2009; **23**: 1044–1049.
8 Han A, Ratner D. What is the role of adjuvant radiotherapy in the treatment of cutaneous squamous cell carcinoma with perineural invasion?. *Cancer* 2007; **109**: 1053–1059.
9 Waxweiler W, Sigmon JR, Sheehan DJ. Adjunctive radiotherapy in the treatment of cutaneous squamous cell carcinoma with perineural invasion. *J Surg Oncol* 2011; **104**: 104–105.
10 Wong J, Breen D, Balogh J *et al.* Treating recurrent cases of squamous cell carcinoma with radiotherapy. *Curr Oncol* 2008; **15**: 229–233.
11 Ch'ng S, Maitra A, Allison RS *et al.* Parotid and cervical nodal status predict prognosis for patients with head and neck metastatic cutaneous squamous cell carcinoma. *J Surg Oncol* 2008; **98**: 101–105.
12 Bessède JP, Vinh D, Khalifa N *et al.* Lymph node metastasis of cutaneous epidermoid carcinomas of the head and neck: prognostic factors and therapeutic strategies. A propos of a series of 13 cases. *Rev Laryngol Otol Rhinol* 2001; **122**: 111–117.
13 Guillot B, Boccara O, Mortier L *et al.* Expert consensus on the management of Merkel cell carcinoma. *Ann Dermatol Venereol* 2011; **138**: 469–474.
14 Jouary T, Leyral C, Dreno B *et al.* Adjuvant prophylactic regional radiotherapy versus observation in stage I Merkel cell carcinoma: a multicentric prospective randomized study. *Ann Oncol* 2011; **23**: 1074–1080.
15 Ghadjar P, Kaanders JH, Poortmans P *et al.* The Essential Role of Radiotherapy in the Treatment of Merkel Cell Carcinoma: A Study from the Rare Cancer Network. *Int J Radiat Oncol Biol Phys* 2011; **81**: e583–e591.
16 Pape E, Rezvoy N, Penel N *et al.* Radiotherapy alone for Merkel cell carcinoma: a comparative and retrospective study of 25 patients. *J Am Acad Dermatol* 2011; **65**: 983–990.
17 Fang LC, Lemos B, Douglas J *et al.* Radiation monotherapy as regional treatment for lymph node-positive Merkel cell carcinoma. *Cancer* 2010; **116**: 1783–1790.
18 Bichakjian CK, Coit DG, Wong SL. Radiation versus resection for Merkel cell carcinoma. *Cancer* 2010; **116**: 1620–1622.
19 Pardo FS, Wang CC, Albert D, Stracher MA. Sebaceous carcinoma of the ocular adnexa: radiotherapeutic management. *Int J Radiat Oncol Biol Phys* 1989; **17**: 643–647.
20 Yen MT, Tse DT, Wu X, Wolfson AH. Radiation therapy for local control of eyelid sebaceous cell carcinoma: report of two cases and review of the literature. *Ophthal Plast Reconstr Surg* 2000; **16**: 211–215.
21 Conill C, Toscas I, Morilla I, Mascaró JM Jr. Radiation therapy as a curative treatment in extraocular sebaceous carcinoma. *Br J Dermatol* 2003; **149**: 441–442.
22 Tomé WA, Jaradat HA, Nelson IA *et al.* Helical tomotherapy: image guidance and adaptive dose guidance. *Front Radiat Ther Oncol* 2007; **40**: 162–178.
23 Mackie TR, Balog J, Ruchala K *et al.* Tomotherapy. *Semin Radiat Oncol* 1999; **9**: 108–117.
24 Beavis AW. Is tomotherapy the future of IMRT? *Br J Radiol* 2004; **77**: 285–295.
25 Welsh JS, Lock M, Harari PM *et al.* Clinical implementation of adaptive helical tomotherapy: a unique approach to image-guided intensity modulated radiotherapy. *Technol Cancer Res Treat* 2006; **5**: 465–479.
26 Dejean C, Kantor G. Henriques de Figueiredo B *et al.* Helical tomotherapy: description and clinical applications. *Bull Cancer* 2010; **97**: 783–789.
27 Kantor G, Mahé MA, Giraud P *et al.* Helical tomotherapy: general methodology for clinical and dosimetric evaluation. *Cancer Radiother* 2006; **10**: 488–491.
28 Kantor G, Mahé MA, Giraud P. French national evaluation for helicoidal tomotherapy: description of indications, dose constraints and set-up margins. *Cancer Radiother* 2007; **11**: 331–337.
29 Murthy V, Master Z, Gupta T *et al.* Helical tomotherapy for head and neck squamous cell carcinoma: dosimetric comparison with linear accelerator-based step-and-shoot IMRT. *J Cancer Res Ther* 2010; **6**: 194–198.
30 Montgomery L, Macpherson M, Gerig L *et al.* Simultaneous treatment of multiple basal cell carcinoma lesions. *Br J Radiol* 2008; **81**: e290–e292.

31 Chatterjee S, Mott JH, Dickson S, Kelly CG. Extensive basal cell carcinoma of the forehead and anterior scalp: use of helical tomotherapy as a radiotherapy treatment modality. *Br J Radiol* 2010; **83**: 538–540.
32 Sterzing F, Neuhof D, Stroebel P *et al.* Sarcomatoid sarcoma of the scalp - helical tomotherapy as a new radiotherapy option. *Eur J Dermatol* 2008; **18**: 731–732.
33 Motta M, Alongi F, De Martin E *et al.* Helical tomotherapy for scalp recurrence of primary eccrine mucinous adenocarcinoma. *Tumori* 2009; **95**: 832–835.
34 Sterzing F, Schubert K, Sroka-Perez G *et al.* Helical tomotherapy. Experiences of the first 150 patients in Heidelberg. *Strahlenther Onkol* 2008; **184**: 8–14.